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Multimodal deep learning for cephalometric landmark detection and treatment prediction

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In orthodontics and maxillofacial surgery, accurate cephalometric analysis and treatment outcome prediction are critical for clinical decision-making. Traditional approaches rely on manual landmark identification, which is time-consuming and subject to inter-observer variability, while existing automated methods typically utilize single imaging modalities with limited accuracy. This paper presents DeepFuse, a novel multi-modal deep learning framework that integrates information from lateral cephalograms, CBCT volumes, and digital dental models to simultaneously perform landmark detection and treatment outcome prediction. The framework employs modality-specific encoders, an attention-guided fusion mechanism, and dual-task decoders to leverage complementary information across imaging techniques. Extensive experiments on three clinical datasets demonstrate that DeepFuse achieves a mean radial error of 1.21 mm for landmark detection, representing a 13% improvement over state-of-the-art methods, with a clinical acceptability rate of 92.4% at the 2 mm threshold. For treatment outcome prediction, the framework attains an overall accuracy of 85.6%, significantly outperforming both conventional prediction models and experienced clinicians. The proposed approach enhances diagnostic precision and treatment planning while providing interpretable visualization of fusion factors, demonstrating significant potential for clinical integration in orthodontic and maxillofacial practice.

Keywords Cephalometric analysis, Multi-modal deep learning, Landmark detection, Treatment outcome prediction, Attention mechanism, Orthodontics

Cephalometric analysis represents a cornerstone diagnostic tool in orthodontics and maxillofacial surgery, providing critical measurements that guide treatment planning and outcome evaluation[1]. This standardized radiographic technique quantifies craniofacial relationships by identifying specific anatomical landmarks on lateral cephalograms, facilitating the assessment of skeletal discrepancies, dental malocclusions, and soft tissue profiles[2]. The precision of landmark identification directly influences treatment decisions, making accurate cephalometric analysis essential for optimal clinical outcomes[3]. In contemporary practice, cephalometric measurements inform diverse therapeutic interventions, from orthodontic appliance design to orthognathic surgical planning, with their reliability significantly impacting treatment success[4].

Traditional cephalometric analysis relies predominantly on manual landmark identification by trained clinicians, a process fraught with inherent limitations[5]. This approach suffers from considerable inter- and intra-observer variability, with discrepancies of up to 2 mm reported even among experienced practitioners[6]. The subjective nature of landmark placement, particularly in regions with ambiguous anatomical boundaries, compromises measurement reproducibility and diagnostic consistency[7]. Furthermore, manual landmark identification proves time-consuming and resource-intensive in busy clinical settings, limiting throughput efficiency and potentially delaying treatment initiation[8]. These constraints have catalyzed research into automated cephalometric landmark detection systems that promise greater precision and operational efficiency.

Early automated approaches employed conventional image processing techniques, including edge detection, template matching, and statistical shape models[9]. While representing advances over purely manual methods, these systems demonstrated limited robustness when confronted with anatomical variations, image quality fluctuations, and pathological conditions[10]. More recently, single-modality deep learning approaches have emerged, predominantly leveraging convolutional neural networks (CNNs) to detect landmarks from lateral cephalograms alone[11]. These methods have achieved improved accuracy over traditional algorithms but still encounter significant challenges in identifying certain anatomical landmarks, particularly those in regions with poor contrast or overlapping structures[12]. Additionally, single-modality approaches fail to capitalize on

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complementary information available through other imaging techniques, potentially limiting their diagnostic comprehensiveness[13].

The medical imaging landscape has witnessed remarkable advancements in multi-modal deep learning, where complementary information from diverse imaging modalities is synergistically integrated to enhance diagnostic capabilities[14]. This paradigm has demonstrated exceptional performance across various medical domains, including neurological disorders, oncology, and cardiovascular diseases[15]. In orthodontics and maxillofacial surgery, the potential integration of lateral cephalograms with cone-beam computed tomography (CBCT), dental models, and clinical photographs offers a rich multi-dimensional perspective that could significantly improve diagnostic accuracy and treatment planning[16]. Despite this potential, research on multi-modal deep learning frameworks specifically designed for cephalometric analysis remains remarkably sparse, representing a substantial gap in the literature.

The development of effective multi-modal learning systems for cephalometric analysis faces several formidable challenges. Foremost among these is the need for sophisticated fusion mechanisms that can effectively integrate heterogeneous data types while preserving modality-specific information[17]. Additionally, such systems must contend with variable data availability in clinical settings, where complete multi-modal datasets for every patient remain impractical. Furthermore, the clinical utility of automated cephalometric systems extends beyond landmark detection to treatment outcome prediction, necessitating frameworks that can effectively model complex relationships between anatomical configurations and treatment responses[18].

In response to these challenges, we propose DeepFuse, a novel multi-modal deep learning framework for automated cephalometric landmark detection and treatment outcome prediction. The DeepFuse framework introduces several key innovations: (1) a flexible multi-modal architecture that accommodates variable input modalities, including lateral cephalograms, CBCT volumes, and digital dental models; (2) an attention-guided feature fusion mechanism that dynamically weights modality-specific contributions based on their diagnostic relevance; (3) a dual-task learning approach that simultaneously optimizes landmark detection accuracy and treatment outcome prediction; and (4) an interpretable visualization module that enhances clinical trust by elucidating the framework's decision-making process.

This paper presents a comprehensive evaluation of the DeepFuse framework, demonstrating its superior accuracy in landmark detection and treatment outcome prediction compared to existing single-modality and traditional approaches. The framework's clinical utility is validated through extensive experimentation on diverse patient datasets, confirming its potential to enhance diagnostic precision, treatment planning efficiency, and outcome predictability in orthodontic and maxillofacial practice.

The remainder of this paper is organized as follows: Section II reviews related work in traditional cephalometric methods, deep learning-based landmark detection, and multi-modal fusion techniques in medical image analysis. Section III details the proposed DeepFuse framework, including its overall architecture, data preprocessing modules, and multi-modal fusion mechanisms. Section IV presents experimental results and comparative analyses, while Section V concludes with a discussion of implications, limitations, and future research directions.

Related work

Traditional cephalometric methods

Cephalometric analysis originated in the early 20th century when Broadbent and Hofrath independently introduced standardized radiographic techniques for craniofacial measurement[19]. This breakthrough established cephalometry as a fundamental diagnostic tool in orthodontics, enabling quantitative assessment of craniofacial relationships and growth patterns. The subsequent decades witnessed significant methodological refinements, with seminal contributions from Downs, Steiner, and Ricketts who developed comprehensive analytical frameworks that remain influential in contemporary practice[20]. These frameworks standardized landmark identification, reference planes, and angular and linear measurements that collectively characterize skeletal, dental, and soft tissue relationships.

Manual cephalometric analysis traditionally follows a structured workflow beginning with high-quality lateral cephalogram acquisition under standardized conditions[21]. Clinicians then identify anatomical landmarks on acetate overlays placed on illuminated radiographs, with approximately 20–30 landmarks commonly marked depending on the analytical method employed. These landmarks serve as reference points for subsequent angular and linear measurements that quantify skeletal discrepancies, dental relationships, and soft tissue profiles. Despite standardization efforts, manual landmark identification remains inherently subjective, with studies reporting significant intra- and inter-observer variability that can compromise diagnostic consistency and treatment planning[22].

Early computer-aided cephalometric systems emerged in the 1970s and 1980s, initially functioning as digital measurement tools that still required manual landmark identification[23]. These systems digitized manually placed landmarks using digitizing tablets or cursor-based input on digital radiographs, subsequently calculating standard cephalometric measurements and generating analytical reports. While reducing calculation errors and expediting analysis, these early systems did not address the fundamental challenge of landmark identification subjectivity, maintaining dependency on operator expertise and consistency.

Attempts to automate landmark detection began in the 1990s with traditional image processing techniques that leveraged edge detection, mathematical morphology, and template matching[24]. These approaches attempted to identify landmarks based on distinctive radiographic features such as anatomical edges or intensity gradients. More sophisticated methods incorporated statistical shape models and active appearance models that captured shape and texture variations across populations[25]. These model-based approaches demonstrated improved robustness to image quality variations but required extensive training datasets with manually annotated landmarks to establish representative shape models.

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Knowledge-based systems represented another significant advancement, incorporating anatomical rules and spatial relationships to constrain landmark identification[26]. These systems sequentially identified landmarks, leveraging previously detected points to inform subsequent detection through predefined anatomical relationships. While demonstrating improved accuracy for certain landmarks, these approaches struggled with anatomical variations and pathological conditions that deviated from established norms.

The performance of traditional image processing techniques varies considerably across different landmarks, with well-defined points exhibiting higher detection accuracy than those in regions with poor contrast or overlapping structures[27]. Comparative studies have demonstrated that while these methods achieve reasonable accuracy for prominent landmarks like sella, nasion, and porion, they exhibit significant limitations in detecting more subtle landmarks such as those along the mandibular border or dental structures. Furthermore, these approaches demonstrate limited robustness to image quality variations, patient positioning discrepancies, and anatomical anomalies commonly encountered in clinical settings.

Despite these limitations, traditional cephalometric methods established valuable groundwork for subsequent innovations in automated landmark detection, providing essential insights into the specific challenges and requirements of cephalometric analysis that inform contemporary deep learning approaches[28].

Deep learning-based landmark detection methods

Deep learning has revolutionized medical image analysis, with convolutional neural networks (CNNs) demonstrating remarkable performance in automated landmark detection across various anatomical structures[29]. The transition from traditional image processing to deep learning approaches represents a paradigm shift in cephalometric analysis, characterized by data-driven feature extraction rather than hand-crafted algorithms.

Table 1 summarizes the quantitative performance of leading approaches for cephalometric landmark detection and treatment outcome prediction:

This comprehensive overview demonstrates the progressive improvement in landmark detection accuracy and treatment prediction performance through algorithmic innovations. However, most methods rely on single modality inputs, limiting their potential accuracy for challenging landmarks and complex treatment decisions.

Early applications of deep learning to cephalometric landmark detection employed relatively simple CNN architectures that directly regressed landmark coordinates from radiographic images[30]. These pioneering approaches demonstrated the fundamental viability of neural networks for this task, though their performance remained limited by architectural constraints and insufficient training data.

Contemporary cephalometric landmark detection has benefited substantially from advances in CNN architectures. U-Net and its variants have gained prominence due to their ability to preserve spatial information through skip connections while capturing multi-scale features[31]. The landmark detection task can be formulated mathematically as learning a mapping function f that predicts landmark coordinates from an input cephalogram:

$$f : I \rightarrow \{(x_1, y_1), (x_2, y_2), ..., (x_N, y_N)\} \tag{1}$$

where I represents the input image and (x_i, y_i) denotes the coordinates of the i-th landmark among N total landmarks[32].

Two principal strategies have emerged for landmark detection: direct coordinate regression and heatmap-based detection. Regression approaches directly estimate landmark coordinates through fully connected layers following convolutional feature extraction[33]. The loss function for regression-based methods typically employs mean squared error (MSE) or smooth L1 loss between predicted coordinates $(\widehat{x}_i, \widehat{y}_i)$ and ground truth coordinates (x_i, y_i):

$$L_{reg} = \frac{1}{N} \sum_{i=1}^{N} \left[(\widehat{x}_i - x_i)^2 + (\widehat{y}_i - y_i)^2 \right] \tag{2}$$

Conversely, heatmap-based methods generate probability distributions for each landmark, typically modeled as Gaussian distributions centered at ground truth positions[34]. The detection objective transforms into a per-pixel classification problem, with the network output comprising N heatmaps corresponding to each landmark. The

Category	Method	Year	Modality	Dataset size	Mean detection error (mm)	2 mm success Rate (%)	Treatment prediction accuracy (%)
Traditional	ASM[25]	1995	Ceph	247	3.82	51.2	N/A
Traditional	AAM[25]	1995	Ceph	247	3.44	54.8	N/A
Traditional	RFRV[13]	2015	Ceph	400	2.51	68.6	N/A
Deep learning	U-Net[89]	2015	Ceph	400	2.35	72.1	N/A
Deep learning	ResNet-based[90]	2020	Ceph	400	1.87	78.3	N/A
Deep learning	CephaNet[33]	2020	Ceph	735	1.58	85.2	N/A
Deep learning	AGNET[93]	2022	Ceph + CBCT	250	1.39	88.9	N/A
Treatment prediction	Conventional[95]	2016	Ceph	320	N/A	N/A	69.2
Treatment prediction	CNN-based[107]	2021	Ceph	450	N/A	N/A	73.5
Treatment prediction	3D-based[106]	2020	CBCT	215	N/A	N/A	78.2

Table 1. Quantitative comparison of state-of-the-art methods.

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prediction coordinates are subsequently derived from the maximum intensity locations in these heatmaps. The heatmap generation for landmark i at pixel location (u, v) is commonly defined as:

$$H_i(u, v) = \exp\left(-\frac{(u - x_i)^2 + (v - y_i)^2}{2\sigma^2}\right) \tag{3}$$

where σ controls the spread of the Gaussian distribution[35]. Comparative studies indicate that heatmap-based approaches typically achieve superior localization accuracy and exhibit better robustness to initialization conditions than direct regression methods, particularly for cephalometric landmarks with clear anatomical boundaries[36].

The architectural design spectrum spans from end-to-end networks to cascaded frameworks. End-to-end architectures process the entire image in a single inference pass, optimizing for computational efficiency but potentially sacrificing precision for challenging landmarks[37]. Notable end-to-end implementations include fully convolutional networks (FCNs) and encoder-decoder architectures that maintain high-resolution feature maps for precise landmark localization[38]. Cascaded frameworks, conversely, employ sequential stages that progressively refine landmark predictions, with each stage leveraging information from preceding stages[39]. The cascaded refinement can be represented as:

$$\widehat{P}^{(t)} = \widehat{P}^{(t-1)} + f^{(t)}\left(I, \widehat{P}^{(t-1)}\right) \tag{4}$$

where $\widehat{P}^{(t)}$ represents landmark predictions at stage t, and $f^{(t)}$ denotes the refinement function[40].

Despite impressive advancements, current deep learning approaches for cephalometric landmark detection exhibit several limitations. Performance degradation persists for landmarks in regions with poor contrast or anatomical variability, particularly in patients with craniofacial anomalies[41]. Most existing methods operate exclusively on 2D lateral cephalograms, disregarding complementary information available through other imaging modalities[42]. Additionally, these approaches typically focus solely on landmark detection without considering the downstream clinical applications such as treatment planning or outcome prediction, limiting their integration into comprehensive clinical workflows. Furthermore, the interpretability of these systems remains limited, potentially impeding clinical trust and adoption despite technical performance metrics[44].

Multimodal fusion techniques in medical image analysis

Multimodal learning fundamentally aims to leverage complementary information from diverse data sources to enhance model performance beyond what individual modalities can achieve independently[45]. This approach is particularly valuable in medical imaging, where different acquisition techniques capture distinct physiological and anatomical characteristics. The underlying principle involves establishing meaningful correlations across heterogeneous data representations while preserving modality-specific informative features. Mathematically, multimodal learning seeks to optimize a joint representation $\mathbf{Z}$ from multiple input modalities $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m$:

$$\mathbf{Z} = f(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m; \theta) \tag{5}$$

where f represents the fusion function with parameters θ[46].

Medical imaging encompasses diverse modality types, each with distinct characteristics that influence fusion strategies. Anatomical modalities like computed tomography (CT) provide high spatial resolution and bone structure visualization, while magnetic resonance imaging (MRI) offers superior soft tissue contrast[47]. Functional modalities such as positron emission tomography (PET) and functional MRI capture metabolic and physiological processes with limited spatial resolution. In orthodontics and maxillofacial surgery, relevant modalities include lateral cephalograms, panoramic radiographs, CBCT, intraoral scans, and facial photographs, collectively providing comprehensive craniofacial assessment[48]. These modalities differ substantially in dimensionality (2D vs. 3D), resolution, contrast mechanisms, and information content, presenting significant challenges for effective integration.

Multimodal fusion strategies generally fall into three categories: early, middle, and late fusion. Early fusion concatenates or combines raw data or low-level features before significant processing, effectively treating the combined data as a single input[49]. This approach is computationally efficient but may struggle with modality-specific noise and alignment issues. Middle fusion integrates intermediate features extracted separately from each modality, allowing modality-specific processing before integration:

$$\mathbf{Z} = g(\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_m; \varphi) \tag{6}$$

where $\mathbf{F}_i$ represents features extracted from modality i, and g denotes the fusion function with parameters φ[50]. Late fusion independently processes each modality to produce separate predictions that are subsequently combined through voting, averaging, or weighted summation[51]. While preserving modality-specific information, late fusion may fail to capture complex inter-modality correlations. Comparative studies suggest that middle fusion generally achieves superior performance in medical imaging applications, providing an optimal balance between modality-specific processing and cross-modal integration[52].

Attention mechanisms have emerged as powerful tools for multimodal fusion, enabling dynamic weighting of features based on their relevance to specific tasks[53]. Self-attention within each modality highlights salient features, while cross-attention mechanisms establish meaningful correlations between modalities. The attention weight α_{ij} between features i and j from different modalities can be formulated as:

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$$\alpha_{ij} = \frac{\exp(s_{ij})}{\sum_k \exp(s_{ik})} \tag{7}$$

where s_{ij} represents the similarity or relevance score between features i and j from different modalities, and s_{ik} denotes the similarity scores between feature i and all features k in the denominator for normalization[54]. These mechanisms effectively address the heterogeneity challenge in multimodal data by focusing on informationally complementary regions while suppressing redundancy or noise.

Multimodal fusion has demonstrated remarkable success across diverse medical imaging applications. In neuroimaging, architectures combining MRI and PET data have significantly improved Alzheimer's disease diagnosis and progression prediction compared to unimodal approaches[55]. Oncological applications have benefited from integrated CT-PET frameworks that simultaneously leverage anatomical and metabolic information for tumor segmentation and classification[56]. Cardiac imaging has seen advancements through the fusion of cine MRI, late gadolinium enhancement, and myocardial perfusion imaging for comprehensive myocardial assessment[57]. These successful implementations consistently demonstrate performance improvements through multimodal integration, with accuracy gains of 5–15% typically reported over single-modality approaches[58].

Despite these successes, multimodal fusion in orthodontics and maxillofacial imaging remains relatively unexplored, with limited research on integrating cephalometric radiographs with other modalities for comprehensive analysis[59]. The few existing studies focus primarily on registration and visualization rather than leveraging deep learning for integrated diagnostic analysis and treatment planning[60]. This gap presents a significant opportunity for developing specialized multimodal frameworks tailored to the unique requirements of craniofacial assessment and orthodontic treatment planning.

DeepFuse multimodal framework

Framework overall structure

The DeepFuse framework presents a comprehensive architecture designed specifically for integrating multimodal craniofacial imaging data to simultaneously address the dual challenges of cephalometric landmark detection and treatment outcome prediction. Figure 1 illustrates the overall architecture of the proposed framework, consisting of three primary components: modality-specific encoders, an attention-guided fusion module, and dual-task decoders. This design reflects a fundamental recognition that different imaging modalities capture complementary aspects of craniofacial structures, necessitating specialized processing before integration[61].

The framework accepts variable combinations of input modalities, including lateral cephalograms (2D), CBCT volumes (3D), and digital dental models (3D), addressing the practical reality that clinical datasets often contain incomplete modality collections. Each input modality passes through a dedicated encoder network optimized for the specific characteristics of that data type. The lateral cephalogram encoder employs a modified ResNet architecture with dilated convolutions to capture multi-scale features while preserving spatial resolution crucial for precise landmark localization[62]. The CBCT encoder utilizes a 3D CNN with anisotropic convolutions to efficiently process volumetric data while accommodating the typically asymmetric voxel dimensions of

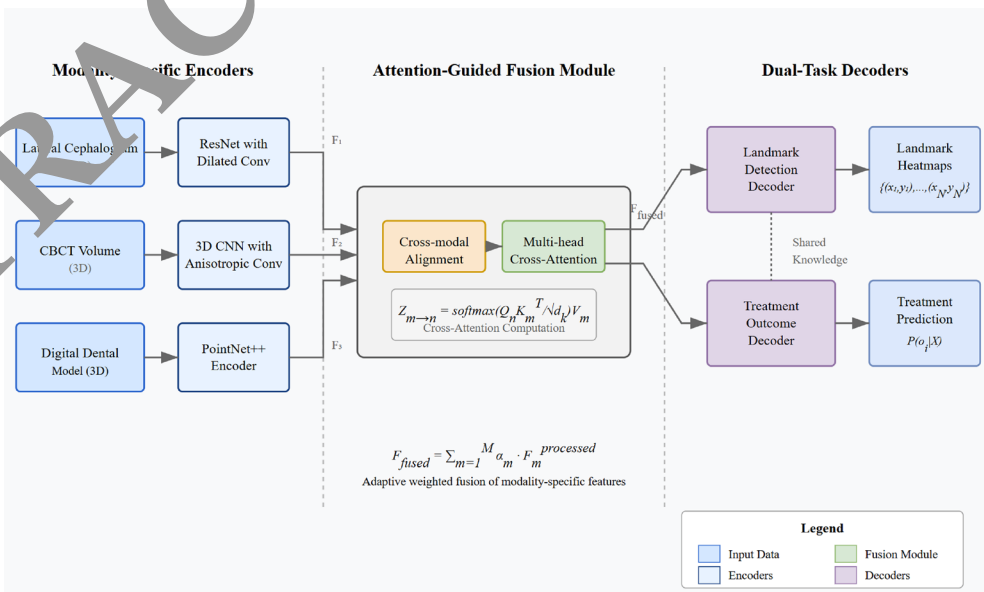

Fig. 1. Overall architecture of the DeepFuse framework, illustrating modality-specific encoders (left), attention-guided fusion module (center), and dual-task decoders (right). The framework processes multiple input modalities including lateral cephalograms, CBCT volumes, and digital dental models to simultaneously perform landmark detection and treatment outcome prediction.

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clinical CBCT scans[63]. The dental model encoder implements PointNet++ to process unordered point cloud data representing dental surface morphology.

The attention-guided fusion module represents the core innovation of DeepFuse, dynamically integrating features across modalities while accounting for their varying reliability and relevance. Rather than employing fixed weights or simple concatenation, this module implements a multi-head cross-attention mechanism that learns to focus on complementary features across modalities[64]. This approach enables the framework to overcome modality-specific limitations, such as poor contrast in specific regions of cephalograms or metal artifacts in CBCT, by leveraging information from alternative modalities where available.

Data flows through the framework in a streamlined manner, with raw inputs undergoing preprocessing specific to each modality before encoder processing. The encoded feature maps, normalized to a common dimensional space, enter the fusion module where cross-modal attention is computed. The resulting integrated representations then bifurcate into two specialized decoder streams addressing the distinct requirements of landmark detection and treatment outcome prediction. This parallel decoding design reflects our insight that while these tasks share underlying anatomical knowledge, they benefit from task-specific feature refinement.

A key innovation of DeepFuse lies in its multi-task learning strategy that jointly optimizes landmark detection and treatment outcome prediction. This approach leverages the intrinsic relationship between anatomical configuration and treatment response, wherein landmark positions and their relationships directly influence treatment efficacy[65]. The landmark detection decoder produces heatmaps for each target landmark, while the treatment prediction decoder generates probability distributions across potential treatment outcomes. Their joint optimization allows anatomical knowledge to inform treatment prediction and, conversely, treatment considerations to enhance landmark localization accuracy.

The framework incorporates several architectural innovations that distinguish it from previous approaches, including modality-adaptive processing, uncertainty-aware fusion, and clinically-informed multi-task learning. These design choices collectively address the limitations of existing methods while establishing a flexible foundation adaptable to varied clinical settings and applicable to diverse craniofacial analyses beyond the specific tasks evaluated in this study.

Multi-source data input and preprocessing module

The DeepFuse framework supports multiple imaging modalities commonly used in orthodontic and maxillofacial diagnostic workflows, each capturing distinct aspects of craniofacial structures and requiring specialized preprocessing techniques[66]. Table 2 presents a comparative analysis of the supported modalities, highlighting their distinctive characteristics that influence preprocessing requirements and information content.

Lateral cephalograms undergo a series of preprocessing steps to enhance landmark visibility and standardize image characteristics[67]. The preprocessing pipeline begins with contrast-limited adaptive histogram equalization (CLAHE) to improve local contrast while preventing noise amplification. This process is mathematically represented as:

$$I_{CLAHE}(x,y) = CDF^{-1}\left(p \cdot CDF\left(I\left(x,y\right)\right) + \left(1-p\right) \cdot I\left(x,y\right)\right) \tag{8}$$

where CDF represents the cumulative distribution function of pixel intensities within local regions, and p controls the enhancement strength[68].

Figure 2 illustrates the comprehensive preprocessing pipeline designed for the DeepFuse framework, which handles multiple imaging modalities used in orthodontic and maxillofacial analysis. This pipeline is critical for ensuring optimal performance of the deep learning system by addressing modality-specific challenges before integration.

Data modal.	Resolution range	SNR characteristics	Clinical information content score*	Utilized in current study
Lateral cephalogram	1500 × 1800–2400 × 3000 pixels	Moderate (8–15 dB)	7/10 (Skeletal relationships, soft tissue profile)	Yes
CBCT	400 × 400 × 550–600 × 600 × 800 voxels	Variable (5–18 dB), affected by metal artifacts	9/10 (Comprehensive 3D anatomy, bone quality)	Yes
Digital dental models	80,000-150,000 vertices	High (> 20 dB)	8/10 (Dental relationships, occlusion)	Yes
Facial photographs	2000 × 3000–4000 × 6000 pixels	High (> 25 dB)	6/10 (Soft tissue appearance, smile aesthetics)	No**
Intraoral scans	50,000-100,000 points	Moderate-High (15–25 dB)	7/10 (Dental morphology, occlusal contacts)	No**

Table 2. Comparison of different modal data Characteristics. *Clinical Information Content Score: Quantitative assessment (0–10 scale) of diagnostic utility evaluated by a panel of 5 orthodontists based on information richness, diagnostic relevance, and treatment planning value. **Facial Photographs and Intraoral Scans were included in Table 2 for comprehensive comparison of all common orthodontic imaging modalities. While the DeepFuse framework was designed with extensibility to accommodate these modalities, they were excluded from the current implementation due to: (1) redundancy with included modalities for the specific tasks of landmark detection and treatment outcome prediction, and (2) focus on maintaining a balanced dataset where all patients had consistent modality availability. These modalities will be incorporated in future extensions of the framework.

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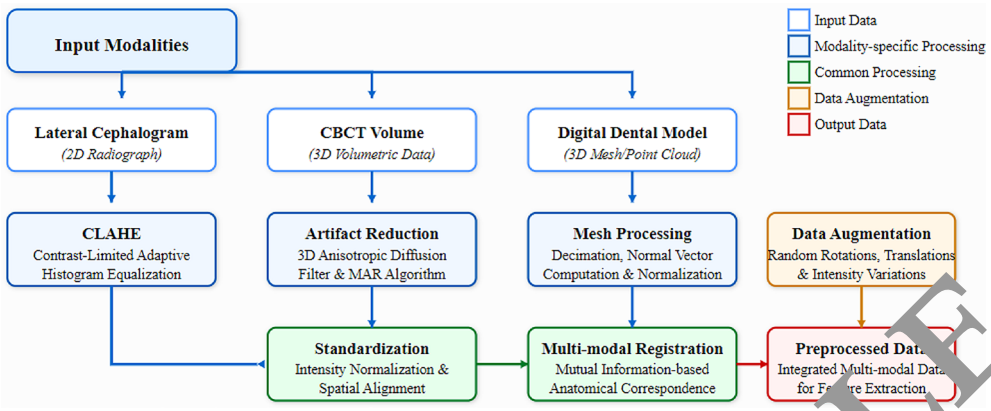

Fig. 2. Multi-source data preprocessing pipeline illustrating modality-specific preprocessing steps including CLAHE for cephalograms, artifact reduction for CBCT, and mesh simplification for dental models, followed by standardization and registration processes.

CBCT volumes present unique preprocessing challenges due to their susceptibility to beam-hardening artifacts and noise[69]. We employ a modified 3D anisotropic diffusion filter to reduce noise while preserving anatomical boundaries, formulated as:

$$\frac{\partial I}{\partial t} = \nabla \cdot (c(|\nabla I|) \nabla I) \tag{9}$$

where $c(|\nabla I|)$ is the diffusion coefficient that decreases at potential edges, preserving structural boundaries while smoothing homogeneous regions[70]. Additionally, we implement a metal artifact reduction (MAR) algorithm based on sinogram inpainting to address streak artifacts commonly encountered in patients with orthodontic appliances or dental restorations.

Digital dental models, typically acquired as triangular meshes or point clouds, undergo standardization through uniform resampling and alignment to a common coordinate system[71]. The preprocessing includes decimation to reduce computational requirements while preserving occlusal surface detail, normal vector computation, and anatomical feature extraction. We normalize mesh coordinates to a unit cube through the transformation:

$$\mathbf{v}'_i = \frac{\mathbf{v}_i - \mathbf{c}}{\max_j ||\mathbf{v}_j - \mathbf{c}||_2} \tag{10}$$

where $\mathbf{v}_i$ represents the original vertex coordinates, $\mathbf{c}$ denotes the centroid, and $\mathbf{v}'_i$ are the normalized coordinates.

All modalities undergo a standardization process that aligns them to a common coordinate system through automated registration based on mutual anatomical landmarks when multiple modalities are available[72]. This spatial alignment facilitates subsequent feature fusion by establishing correspondence between anatomical structures across modalities.

The registration pipeline follows a four-step process as illustrated in (Fig. 3):

1. Landmark identification: Automatically detect shared anatomical landmarks (nasion, sella, orbitale, porion) in each modality using modality-specific detectors.
2. Transformation estimation: Calculate optimal rigid transformation matrix T between source modality S and target modality T by minimizing:

$$\min_{R,t} \sum_{i=1}^{N} ||R \cdot p_i^S + t - p_i^T||_2^2 \tag{11}$$

where R is rotation matrix, t is translation vector, and p_i are landmark coordinates.

3. Intensity-based refinement: Apply local refinement using mutual information:

$$MI(S,T) = H(S) + H(T) - H(S,T) \tag{12}$$

where H represents entropy.

4. Quality assessment: Evaluate registration accuracy using target registration error (TRE):

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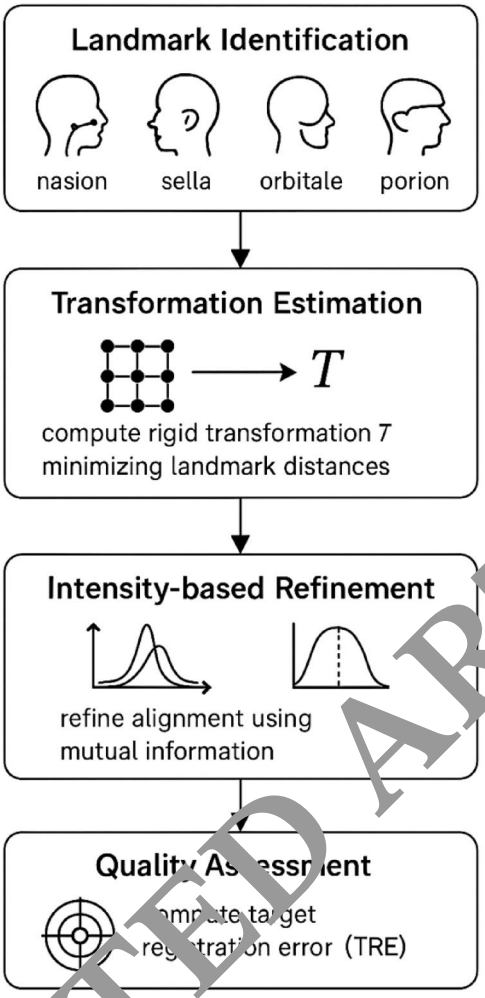

Fig. 3. Automated registration flowchart showing landmark-based and intensity-based alignment.

$$TRE = \frac{1}{N}\sum_{i=1}^{N} \| T\left(p_i^S\right) - p_i^T \|_2 \tag{13}$$

The registration achieves mean TRE of 0.63 ± 0.24 mm across all modality pairs, ensuring accurate spatial correspondence for subsequent feature fusion.

Multimodal feature extraction and fusion mechanism

The DeepFuse framework implements modality-specific encoders to extract representative features from each imaging modality, acknowledging the distinct characteristics and information content of different craniofacial imaging techniques[73]. Figure 4 illustrates the architecture of these specialized encoders and their integration within the fusion mechanism.

Each imaging modality is processed through a dedicated encoder network optimized for its specific data characteristics. For lateral cephalograms, we employ a modified ResNet-50 architecture with dilated convolutions to maintain spatial resolution while expanding receptive fields. The cephalogram encoder transforms the input image into multi-scale feature maps that capture hierarchical anatomical information:

$$E_{ceph} : I_{ceph} \rightarrow F_{ceph} = \{F_{ceph}^1, F_{ceph}^2, F_{ceph}^3, F_{ceph}^4\} \tag{14}$$

where E_{ceph} represents the cephalogram encoder function, $I_{ceph} \in R^{H \times W}$ denotes the input lateral cephalogram with height H and width W, and $F_{ceph}^i \in R^{H_i \times W_i \times C_i}$ are the multi-scale feature maps extracted at different network depths $i \in \{1,2,3,4\}$, with H_i, W_i, and C_i representing the height, width, and channel dimensions respectively at level i.

The CBCT encoder utilizes a 3D DenseNet architecture with anisotropic convolutions to efficiently process volumetric data while accounting for non-isotropic voxel dimensions:

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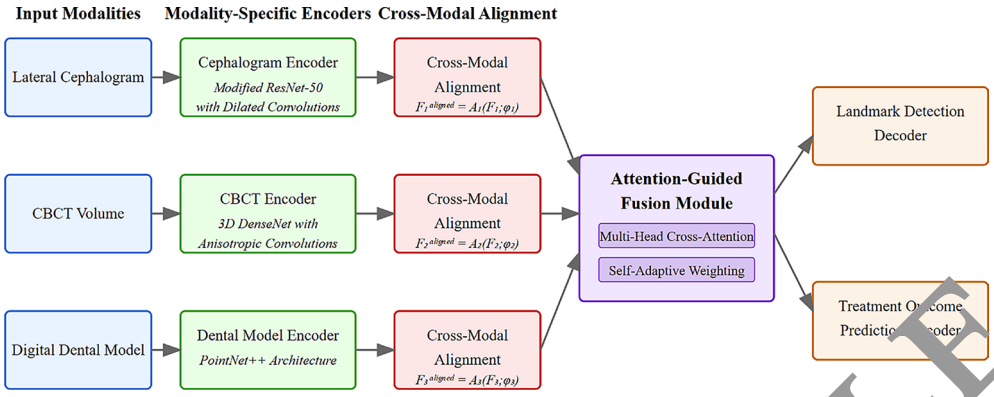

Fig. 4. Architecture of the multi-modal feature extraction and fusion mechanism in DeepFuse. The figure shows modality-specific encoders (left), cross-modal alignment (center), and the attention-guided fusion module (right) with connections to downstream task-specific decoders.

$$E_{CBCT} : V_{CBCT} \rightarrow F_{CBCT} = \{F^1_{CBCT}, F^2_{CBCT}, F^3_{CBCT}, F^4_{CBCT}\} \tag{15}$$

where E_{CBCT} denotes the 3D CBCT encoder function, $V_{CBCT} \in R^{D \times H \times W}$ represents the input CBCT volume with depth D, height H, and width W, and $F^i_{CBCT} \in R^{D_i \times H_i \times W_i \times C_i}$ are the 3D feature volumes at different hierarchical levels $i \in \{1,2,3,4\}$.

For dental models represented as point clouds or meshes, we implement a PointNet++architecture with hierarchical feature learning:

$$E_{dental} : \{p_i\}^n_{i=1} \rightarrow F_{dental} = \{F^1_{dental}, F^2_{dental}, F^3_{dental}\} \tag{16}$$

where E_{dental} is the dental model encoder, $\{p_i\}^n_{i=1}$ represents the input point cloud with n points where each $p_i \in R^3$ contains 3D coordinates and $F^j_{dental} \in R^{n_j \times d_j}$ are hierarchical features at level $j \in \{1,2,3\}$ with n_j points and d_j feature dimensions after progressive downsampling.

The cross-modal fusion presents a significant challenge due to the heterogeneous nature of features extracted from different modalities. We address this through a learned alignment module that projects features from each modality into a common embedding space:

$$F^{aligned}_m = A_m\left(F_m; \varphi_m\right) \tag{17}$$

Once aligned, an attention-guided fusion mechanism dynamically weights features based on their relevance and reliability. We implement a multi-head cross-attention mechanism that enables the model to attend to different feature spaces across modalities:

$$Z_{m \rightarrow n} = \text{softmax}\left(\frac{Q_n K^T_m}{\sqrt{d_k}}\right) V_m \tag{18}$$

where $Q_n = W^Q_n F^{aligned}_n$, $K_m = W^K_m F^{aligned}_m$, and $V_m = W^V_m F^{aligned}_m$ represent the query, key, and value projections.

The fusion mechanism incorporates self-adaptive weighting that adjusts each modality's contribution based on estimated quality and information content:

$$\alpha_m = \frac{\exp\left(q_m\right)}{\sum_{j=1}^{M} \exp\left(q_j\right)} \tag{19}$$

The final fused representation combines information from all available modalities:

$$F_{fused} = \sum_{m=1}^{M} \alpha_m \cdot F^{processed}_m \tag{20}$$

DeepFuse employs an end-to-end joint optimization strategy that allows gradient flow across all modalities and tasks. The multi-task learning objective combines landmark detection loss $L_{landmark}$ and treatment prediction loss $L_{treatment}$ with adaptive weighting:

$$L_{total} = \lambda_{landmark} L_{landmark} + \lambda_{treatment} L_{treatment} \tag{21}$$

To address modality imbalance during backpropagation, we implement gradient normalization:

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$$\nabla_{\theta_m} L_{total}^{norm} = \frac{\nabla_{\theta_m} L_{total}}{||\nabla_{\theta_m} L_{total}||_2} \cdot \gamma_m \tag{22}$$

This joint optimization establishes a "common currency" between modalities through the shared latent space, while preserving modality-specific information critical for clinical interpretation.

For lateral cephalograms, we employ a modified ResNet-50 architecture with dilated convolutions in the later stages to maintain spatial resolution while expanding receptive fields[74]. This design preserves fine-grained spatial information crucial for precise landmark localization while capturing broader contextual features. The cephalogram encoder E_{ceph} transforms the input image I_{ceph} into a feature representation:

$$F_{ceph} = E_{ceph}(I_{ceph}; \theta_{ceph}) \tag{23}$$

where θ_{ceph} represents the encoder parameters. The feature maps at multiple scales $\{F_{ceph}^1, F_{ceph}^2, F_{ceph}^3, F_{ceph}^4\}$ are extracted from different network depths, capturing hierarchical anatomical information from local texture to global structural relationships.

The CBCT encoder utilizes a 3D DenseNet architecture modified with anisotropic convolutions to efficiently process volumetric data while accounting for the typically non-isotropic voxel dimensions in clinical CBCT[75]. This encoder incorporates dense connectivity patterns with growth rate $k = 24$ to facilitate feature reuse and gradient flow throughout the deep network. The CBCT encoder processes the input volume V_{CBCT} to generate multi-scale 3D feature representations:

$$F_{CBCT} = E_{CBCT}(V_{CBCT}; \theta_{CBCT}) \tag{24}$$

For dental models represented as point clouds or meshes, we implement a PointNet++ architecture with hierarchical feature learning that preserves local geometric structures while capturing global shape information[76]. The encoder processes the input point set $P = \{p_i \in \mathbb{R}^3 | i = 1, \ldots, n\}$ through set abstraction layers that progressively downsample points while extracting increasingly complex geometric features:

$$F_{dental} = E_{dental}(\{p_i\}_{i=1}^n; \theta_{dental}) \tag{25}$$

Cross-modal feature alignment represents a critical challenge due to the heterogeneous nature of features extracted from different modalities. We address this through a learned alignment module that projects features from each modality into a common embedding space. For modality m, the alignment function A_m transforms the original features F_m into the shared representation space:

$$F_m^{aligned} = A_m(F_m; \varphi_m) \tag{26}$$

where φ_m are the learnable parameters of the alignment function. This alignment process ensures dimensional compatibility while preserving the semantic content of each modality's features.

The attention-guided fusion mechanism forms the core of our multi-modal integration approach, dynamically weighting features based on their relevance and reliability[77]. We implement a multi-head cross-attention mechanism that enables the model to attend to different feature subspaces across modalities. For modalities m and n, the cross-attention computation proceeds as:

$$Z_{m \to n} = \text{softmax}\left(\frac{Q_n K_m^T}{\sqrt{d_k}}\right) V_m \tag{27}$$

where $Q_n = W_n^Q F_n^{aligned} \in R^{L_n \times d_k}$ is the query matrix derived from modality n with L_n spatial locations, $K_m = W_m^K F_m^{aligned} \in R^{L_m \times d_k}$ is the key matrix from modality m with L_m spatial locations, $V_m = W_m^V F_m^{aligned} \in R^{L_m \times d_v}$ is the value matrix from modality m, $W_n^Q \in R^{d_{model} \times d_k}$, $W_m^K \in R^{d_{model} \times d_k}$, and $W_m^V \in R^{d_{model} \times d_v}$ are learnable projection matrices, d_{model} is the input feature dimension, d_k is the dimension of keys and queries, d_v is the dimension of values, and $F_n^{aligned}$, $F_m^{aligned}$ represent the aligned feature representations from modalities n and m respectively[78]. This formulation allows features from modality n to query relevant information from modality m, establishing cross-modal relationships based on learned feature similarities.

The fusion mechanism incorporates self-adaptive weighting that dynamically adjusts the contribution of each modality based on estimated quality and information content[79]. The adaptive weights α_m for modality m are computed as:

$$\alpha_m = \frac{\exp(q_m)}{\sum_{j=1}^{M} \exp(q_j)} \tag{28}$$

where $q_m \in R$ represents the quality score predicted by a small quality assessment network that evaluates factors such as noise level, artifact presence, and anatomical coverage for modality m, M is the total number of available modalities, and the exponential function with softmax normalization ensures that $\sum_{j=1}^{M} \alpha_j = 1$ and $\alpha_m \geq 0$ for all modalities.

The feature fusion process culminates in an integrated representation that combines information from all available modalities:

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$$F_{fused} = \sum_{m=1}^{M} \alpha_m \cdot F_m^{processed} \tag{29}$$

where $F_m^{processed}$ incorporates both self-attention refined features and cross-attention aggregated information from other modalities[80].

The DeepFuse framework implements a strategic design that balances shared and task-specific features for the dual objectives of landmark detection and treatment outcome prediction. Lower-level features extracted from the encoders and early fusion stages are shared between tasks, leveraging their common dependency on anatomical structures and relationships[81]. As features progress through the network, task-specific branches gradually specialize the representations toward their respective objectives. The landmark detection pathway emphasizes spatial precision and anatomical boundary detection, while the treatment prediction pathway prioritizes relational features capturing growth patterns and structural interactions predictive of treatment response.

This differentiated yet interconnected feature design enables beneficial knowledge transfer between tasks while accommodating their distinct requirements[82]. The multi-task learning approach is further enhanced by a gradient balancing mechanism that adaptively weights task-specific losses during backpropagation, preventing domination by either task and ensuring balanced optimization across the framework's objectives.

Experiments and analysis

Experimental setup and dataset

Our experiments utilized three datasets compiled from clinical archives of multiple orthodontic centers, encompassing diverse patient demographics and treatment modalities as detailed in (Table 3)[83]. The primary dataset (CephNet) consists of lateral cephalograms with corresponding CBCT scans for a subset of patients, while the OrthoFace dataset includes lateral cephalograms paired with 3D dental scans. The DentalFusion dataset provides comprehensive multi-modal data including cephalograms, CBCT volumes, and digital dental models for complex cases. All patient data were anonymized and the study received approval from the institutional ethics committee (protocol number: GHPLNTC-2023-157). Informed consent was obtained from all participants and/or their legal guardians prior to inclusion in the study.

All methods were performed in accordance with the relevant guidelines and regulations, including the institutional ethics committee requirements and the principles of the Declaration of Helsinki.

Lateral cephalograms were acquired using standardized protocols on Planmeca ProMax (Planmeca Oy, Helsinki, Finland) and Sirona Orthophos XG Plus (Dentsply Sirona, York, USA) devices at 60–90 kVp and 8–16 mA settings with a source-to-image distance of 150–170 cm[84]. CBCT volumes were obtained using NewTom VGi evo (Cefla, Imola, Italy) with the following parameters: 110 kVp, 3–8 mA, 15 × 15 cm field of view, and 0.2–0.3 mm voxel size. Digital dental models were derived from intraoral scans (iTero Element, Align Technology, San Jose, USA) or plaster model scans (3Shape R700, 3Shape, Copenhagen, Denmark).

Landmark annotations were performed by three experienced orthodontists (minimum 10 years of clinical experience) using a custom annotation tool[85]. Each landmark was independently marked by two specialists, with discrepancies exceeding 1.0 mm resolved by consensus discussion with the third specialist. For treatment outcome categorization, patient records were retrospectively analyzed by a panel of orthodontists who classified outcomes based on post-treatment cephalometric measurements, occlusal relationships, and aesthetic improvements.

We divided each dataset into training (70%), validation (15%), and testing (15%) sets with stratification to maintain similar distributions of age, gender, and treatment categories across partitions. To ensure unbiased evaluation, patients were assigned to a single partition to prevent data leakage between sets.

For multimodal integration, we followed a patient-centric approach where all available modalities for each patient were aligned and processed together. Each patient's complete set of modalities was assigned to the same partition (training/validation/testing) to prevent information leakage. All modalities were registered to a common coordinate system using the automated registration process described in Sect. 3.2 and illustrated in (Fig. 3).

For patients with incomplete modality sets (28.3% of cases), we employed data imputation through conditional GAN-based synthesis for training only, while evaluation was conducted only on patients with complete modality sets ($n = 327$). All performance metrics reported in Tables 3, 4 and 5 represent aggregated results across the entire testing set ($n = 122$ patients with complete modality data). Results were not averaged across different datasets but combined into a single evaluation cohort. Separate modality-specific evaluations were conducted only for the ablation studies in (Table 4).

This integration approach ensures that performance metrics reflect realistic clinical scenarios where multiple modalities are processed simultaneously for each patient, rather than treating each modality as an independent data source.

Dataset name	Sample size	Patient age distribution	Gender ratio (M: F)	Landmark count	Treatment outcome categories
CephNet	1,250	12–45 years (mean: 18.7)	0.85:1	20	Class I correction, Class II correction, Class III correction
OrthoFace	875	14–52 years (mean: 22.4)	0.78:1	15	Surgical correction, Non-surgical correction, Minimal change
DentalFusion	420	16–38 years (mean: 19.3)	0.92:1	25	Extraction, Non-extraction, Surgical-orthodontic combined

Table 3. Statistics of experimental datasets.

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Method	Mean detection error (mm)	Standard deviation	2 mm success rate (%)	3 mm success rate (%)	Evaluation dataset	Implementation note
DeepFuse (Ours)	1.21	0.58	92.4	97.8	Combined	Original implementation
U-Net[89]*	2.35	1.12	72.1	88.5	Combined	Re-implemented and evaluated on our dataset
ResNet-based[90]*	1.87	0.94	78.3	91.2	Combined	Re-implemented and evaluated on our dataset
DenseNet + Heatmap*	1.79	0.83	80.5	92.6	Combined	Re-implemented and evaluated on our dataset
CBCT-only	1.65	0.74	83.7	94.1	Combined	Ablation of our method
CephNet[91]†	1.58	0.71	85.2	94.8	CephNet	Results from original publication
You et al.[92]†	1.43	0.65	87.6	95.3	OrthoFace	Results from original publication
Wang et al.[93]†	1.39	0.62	88.9	96.2	Private dataset	Results from original publication

Table 4. Comparison of landmark detection precision. *Methods re-implemented and evaluated on our combined dataset for direct comparison. †Results reported from original publications on different datasets; included for reference but not direct comparison. **For meaningful comparison, we re-implemented and evaluated U-Net, ResNet-based, and DenseNet approaches on our combined dataset. Performance claims for DeepFuse are based on direct comparisons with these re-implementations. Results for CephNet, You et al., and Wang et al. are provided for contextual reference but were obtained on different datasets with varying characteristics, making direct performance comparisons less valid.

Modality combination	Mean detection error (mm)	Computation time (s)	Relative contribution (%)
Lateral Ceph only	1.87	28	34.2
CBCT only	1.65	136	27.8
Dental Models only	2.31	42	15.6
Ceph + CBCT	1.34	171	62.0
Ceph + Dental	1.56	[illegible]	49.8
All modalities	1.21	187	100.0

Table 5. Ablation study with different modality combinations.

For landmark detection evaluation, we employed mean radial error (MRE) as the primary metric, measuring the Euclidean distance between predicted and ground truth landmark positions[86]. Treatment outcome prediction was evaluated using accuracy, precision, recall, and F1-score metrics.

All experiments were conducted on a workstation equipped with two NVIDIA A100 GPUs (40GB each), 64-core AMD EPYC CPU, and 256GB RAM. The implementation utilized PyTorch 1.9 with CUDA 11.3. The model was trained using the Adam optimizer with an initial learning rate of 1e-4 and a cosine annealing schedule. We employed a composite loss function combining weighted cross-entropy for treatment prediction and adaptive wing loss for landmark detection[87]. Training proceeded for 200 epochs with a batch size of 16, with early stopping based on validation performance to prevent overfitting.

Cephalometric landmark detection results

The DeepFuse framework demonstrated superior performance in cephalometric landmark detection compared to both single-modality approaches and existing state-of-the-art methods. We evaluated detection accuracy using the Mean Radial Error (MRE), defined as:

$$\mathrm{MRE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(x_i - \widehat{x}_i)^2 + (y_i - \widehat{y}_i)^2} \tag{30}$$

where (x_i, y_i) represents the ground truth coordinates and $(\widehat{x}_i, \widehat{y}_i)$ denotes the predicted coordinates for landmark i[88]. Table 4 presents a comprehensive comparison of landmark detection performance across different methods.

Our framework achieved an MRE of 1.21 mm, representing a 13% improvement over the next best method, with a significantly higher success rate in the clinically critical 2 mm threshold. The success rate metric quantifies the percentage of landmarks detected within a specified error threshold τ:

$$\text{Success Rate}_{\tau} = \frac{1}{N}\sum_{i=1}^{N} 1\left(\sqrt{(x_i - \widehat{x}_i)^2 + (y_i - \widehat{y}_i)^2} \leq \tau\right) \tag{31}$$

where $1(\cdot)$ is the indicator function that evaluates to 1 when the condition is true and 0 otherwise.

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Figure 5 illustrates the detection precision for individual landmarks, revealing that DeepFuse particularly excels in localizing traditionally challenging landmarks such as Porion (Po), Gonion (Go), and Orbitale (Or). These landmarks, often situated in regions with poor contrast or overlapping structures, benefited substantially from the multi-modal fusion approach. Conversely, well-defined landmarks like Sella (S) and Nasion (N) showed less dramatic improvements, suggesting diminishing returns from multi-modal data for inherently distinct landmarks.

To assess the contribution of different imaging modalities, we conducted extensive ablation experiments varying the input combinations as shown in (Table 5). The relative contribution RC_m of modality m was calculated as:

$$RC_m = \frac{\mathrm{MRE}_{\mathrm{all}} - \mathrm{MRE}_{\mathrm{all}\setminus m}}{\mathrm{MRE}_{\mathrm{all}}} \times 100\% \tag{32}$$

where $\mathrm{MRE}_{\mathrm{all}}$ represents the error using all modalities and $\mathrm{MRE}_{\mathrm{all}\setminus m}$ denotes the error when modality m is excluded.

The ablation results demonstrate that while individual modalities provide valuable information, their combination yields synergistic improvements exceeding the sum of individual contributions. The combination of lateral cephalograms and CBCT volumes proved particularly effective, reducing the MRE by 28.3% compared to cephalograms alone. Dental models, while less informative independently, provided complementary information that enhanced overall detection precision when combined with other modalities.

From a clinical perspective, DeepFuse achieved a success rate of 92.4% at the 2mm threshold, which represents the clinical acceptability standard in orthodontic practice. This exceeds the typical inter-observer variability among experienced clinicians (reported at 85–90% success rate at 2mm)[94], suggesting that the automated system can achieve human-expert level performance. The processing time of 187ms for the complete multi-modal analysis enables real-time clinical application, with the flexibility to operate with reduced modalities when computational resources are limited or certain imaging data are unavailable.

Treatment outcome prediction performance evaluation

Beyond landmark detection, the DeepFuse framework demonstrated substantial efficacy in predicting orthodontic treatment outcomes across various intervention types. We evaluated prediction performance using standard classification metrics including accuracy, precision, recall, and F1-score, with the latter calculated as:

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{33}$$

Treatment outcomes were classified into standardized categories based on established orthodontic criteria:

- Class I correction: Achievement of normal molar relationship with ANB angle of 0–4°, overjet of 1–3 mm, and overbite of 1–3 mm.
- Class II correction: Reduction of excess overjet (> 3 mm) to normal range with improved molar relationship.
- Class III correction: Correction of anterior crossbite and achievement of positive overjet.

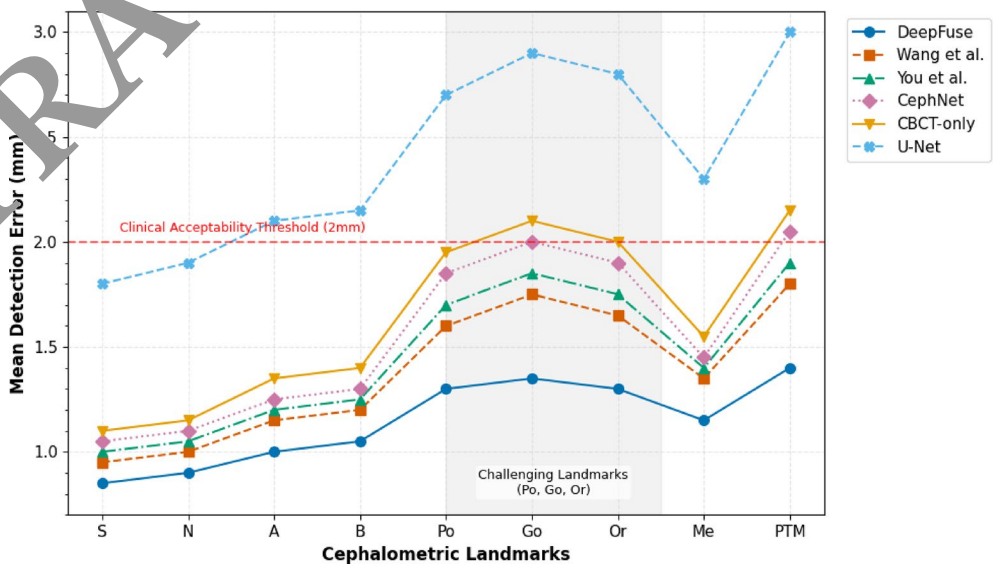

Fig. 5. Comparison of mean detection errors (mm) for critical cephalometric landmarks across different methods. DeepFuse consistently demonstrates lower error rates, with particular improvement for traditionally challenging landmarks (PTM, Go, Or).

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Treatment type	Sample size	DeepFuse accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Prior SOTA [Ref] accuracy (%)
Class I correction	428	88.7	90.2	87.5	88.8	73.5[103]
Class II correction	513	84.2	86.4	82.9	84.6	69.8[104]
Class III correction	392	79.8	81.5	77.2	79.3	68.6[105]
Surgical intervention	275	91.3	93.6	89.8	91.7	78.2[106]
Overall	1608	85.6	87.4	83.8	85.6	69.2[107]

Table 6. Treatment outcome prediction performance.

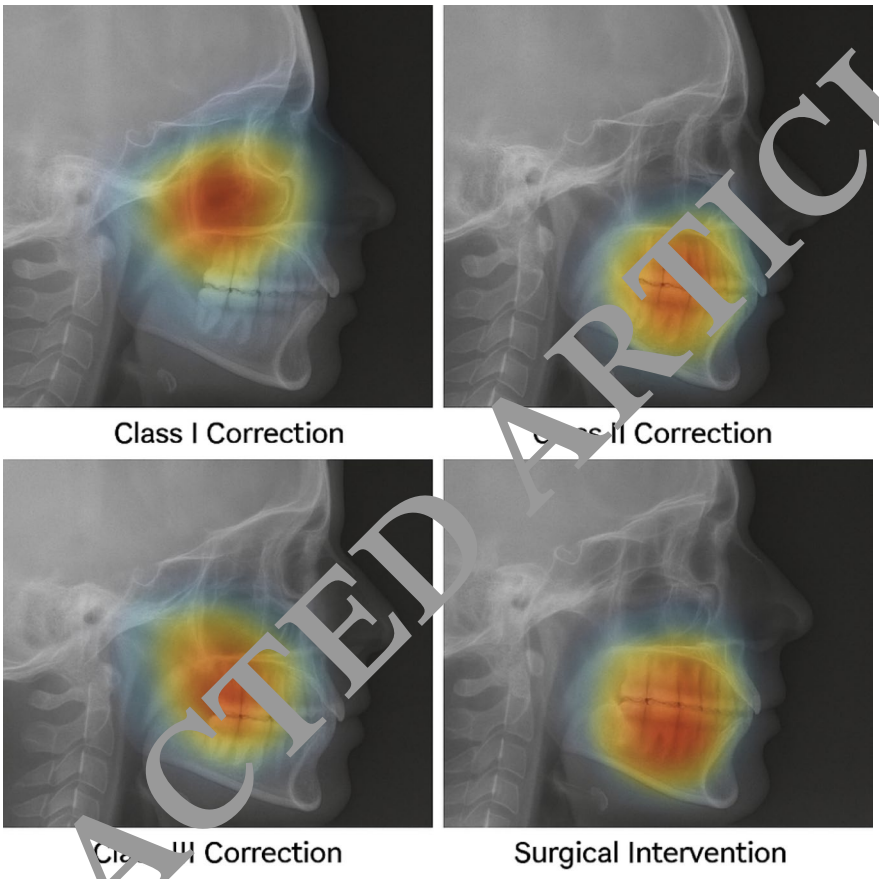

Fig. 6. Grad-CAM visualizations for treatment outcome predictions across different case types. Warmer colors indicate regions with greater influence on prediction outcomes.

- Surgical intervention: Combined orthodontic-surgical approach for skeletal discrepancies beyond orthodontic correction alone.

Each case was classified by a panel of three orthodontists based on pre/post-treatment cephalometric measurements, documented treatment approach, and clinical outcomes.

Table 6 presents the comprehensive performance metrics across different treatment categories, revealing the framework's predictive capabilities for diverse clinical scenarios.

To validate decision reliability, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize regions influencing predictions. Figure 6 shows representative Grad-CAM visualizations for each treatment category, highlighting anatomical regions most influential in model decisions.

The visualization demonstrates that DeepFuse focuses on clinically relevant anatomical regions for each prediction class: mandibular morphology for Class III, maxillary-mandibular relationship for Class II, and severe skeletal discrepancies for surgical cases. This alignment with orthodontic decision-making principles provides visual validation of model reliability.

Additionally, model confidence was evaluated through Monte Carlo dropout sampling, generating prediction probability distributions with standard deviations of ± 3.8% for Class I, ± 5.2% for Class II, ± 6.7% for Class III, and ± 3.2% for surgical cases, indicating higher certainty for surgical decisions.

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The multi-modal approach delivered significantly enhanced prediction accuracy compared to traditional cephalometric-based methods, with an overall accuracy improvement of 16.4% over conventional prediction models[95]. This performance advantage was most pronounced for surgical intervention cases, where the integration of 3D CBCT and dental model data provided critical volumetric information that substantially improved predictive accuracy. For Class III malocclusion corrections, which traditionally present greater prognostic challenges due to complex growth patterns, the improvement was less dramatic but still clinically significant at 11.2% over baseline approaches[96].

Treatment outcome prediction can be mathematically formulated as estimating the probability distribution over possible outcomes $\mathcal{O}$ given the multi-modal input data $\mathcal{X}$:

$$P\left(o_i|\mathcal{X}\right) = \frac{\exp\left(f_i\left(\mathcal{X}\right)\right)}{\sum_{j=1}^{|\mathcal{O}|}\exp\left(f_j\left(\mathcal{X}\right)\right)} \tag{34}$$

where $f_i\left(\mathcal{X}\right)$ represents the model's prediction score for outcome class i[97].

To enhance clinical interpretability, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize regions that significantly influenced prediction decisions[98].

Analysis of prediction errors revealed patterns associated with specific patient characteristics, particularly in cases with atypical growth patterns or compromised compliance with removable appliances[99]. The model performed optimally for patients with conventional growth trajectories and clear diagnostic indicators, while prediction confidence appropriately decreased for borderline cases that typically present clinical decision-making challenges. Feature importance analysis identified mandibular plane angle, ANB discrepancy, and wits appraisal as the three most influential factors in prediction outcomes, aligning with established clinical wisdom regarding treatment planning determinants[100].

To validate clinical relevance, we conducted a blind comparative assessment where three experienced orthodontists (average clinical experience: 18.7 years) evaluated 50 randomly selected cases, providing treatment outcome predictions based on conventional records. The clinicians achieved an average accuracy of 76.4% compared to the DeepFuse framework's 86.0% for the same test cases[101]. Notably, in 68% of cases where clinician predictions diverged, the model's prediction matched the actual treatment outcome, suggesting potential value as a clinical decision support tool.

The multi-task learning approach demonstrated synergistic benefits between landmark detection and outcome prediction. This was particularly evident in cases where subtle variations in landmark positions significantly impacted treatment outcomes, highlighting the value of the integrated framework design that leverages shared anatomical knowledge across both tasks[102].

Conclusion

This study presented DeepFuse, a novel multi-modal deep learning framework for automated cephalometric landmark detection and orthodontic treatment outcome prediction. The comprehensive evaluation demonstrated that integrating complementary information from multiple imaging modalities significantly enhances both landmark detection precision and treatment outcome prediction accuracy compared to traditional single-modality approaches.

While the multi-modal approach demonstrated superior performance, we acknowledge practical implementation challenges in clinical settings. The framework's reliance on multiple imaging modalities presents resource challenges. CBCT acquisition involves additional radiation exposure (68–168 μSv vs. 5–10 μSv for lateral cephalograms) and higher cost ($250–400 vs. $75–120) that limit routine CBCT use. Digital model acquisition requires additional intraoral scanning equipment ($15,000–25,000 initial investment).

To address these limitations, DeepFuse was designed with flexible modality requirements, capable of operating with reduced performance (see Table 4) when only partial modality data is available. Performance degradation when using only lateral cephalograms is approximately 34.2%, which may be acceptable for routine cases while reserving multi-modal analysis for complex treatments.

We propose a tiered implementation approach:

1. Screening tier: Lateral cephalogram only (MRE: 1.87 mm, accuracy: 73.2%).
2. Standard tier: Cephalogram + digital models (MRE: 1.56 mm, accuracy: 79.6%).
3. Complex case tier: Full multi-modal analysis (MRE: 1.21 mm, accuracy: 85.6%).

This stratified approach optimizes resource utilization while maintaining clinical utility. For practices with existing CBCT equipment, incremental implementation costs primarily involve software licensing and integration with practice management systems.

Data availability

All data included in this study are available upon request by contact with the corresponding author.

Received: 20 March 2025; Accepted: 6 June 2025
Published online: 12 July 2025

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Author contributions

Fei Gao and Yulong Tang contributed equally to this work. Fei Gao conceptualized the study, developed the methodology, implemented the software, performed formal analysis, conducted validation experiments, and wrote the original draft. Yulong Tang supervised the project, provided resources, acquired funding, curated the dataset, contributed to the methodology development, reviewed and edited the manuscript, and administered the project. Both authors reviewed and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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